

Society V1

Anatomy of the 42

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The 42 conceptual variables that compose Society V1 represent abstractions of the fundamental forces operating within society. These concepts (e.g., DIVERSITY , POVERTY , MENTAL HEALTH , etc.) are mathematically defined by an influence matrix, a collection of output edges with weighted polarity expressed as normalized values between -1 and 1. Each variable is materially tethered to a physical slider on the Society V1 controller board, allowing users to interact directly with the model.

Changing the value of a variable by moving its corresponding slider adjusts the directional intensity of the influence it exerts on others connected to it. No change occurs in isolation; rather, it incites conditional responses along associated pathways, dynamically reshaping the collective state of the system.

To illustrate: If you change the value of LAW ENFORCEMENT , depending on the current vector state of the system, CRIME will change its state by an amount determined by the weighted edge connecting the two variables. When CRIME settles into its new position, its own network of influences will trigger, affecting downstream variables like DEATH RATE and PUBLIC TRUST .

The mechanism may appear straightforward, but it naturally raises important questions. How are the weights assigned? What justifications or rationale support the exact values of these weights? Are they speculative? Is it arbitrary? How can a specific value be deduced from empirical research when the findings themselves often present convoluted observations or abstract outcomes inferred by researchers?

This document outlines how the values for these concepts are informed by and inferred from aggregated research findings, comparative data, and disciplined observational reasoning.

For readers interested in the broader overview of the system, see *Society V1: Whitepaper*. For in-depth technical implementation, including the influence engine and mathematical foundations, see *Computational & Mathematical Architecture (CMA)*.

It should be underscored that these values, along with the mathematics directing them, are ultimately transcribed by a single mortal mind¹ attempting, through large bodies of research and literature, to painstakingly quantify abstractions far more complex than any model can fully capture. The act of bounding these broad abstractions into discrete values inevitably reduces their individual richness; however, at scale, when enough are structured together, this reduction becomes the very mechanism that makes modeling society possible.

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|----------------------|------------------------|---------------------|----------------------|
| 1. Addiction | 12. Education Cost | 23. Homelessness | 34. Poverty |
| 2. Agriculture | 13. Education Quality | 24. Immigration | 35. Public Trust |
| 3. Authoritarianism | 14. Employment Rate | 25. Infrastructure | 36. Rule of Law |
| 4. Birthrate | 15. Energy Production | 26. Innovation | 37. Spirituality/God |
| 5. Corruption | 16. Equality | 27. Law Enforcement | 38. Sickness/Disease |
| 6. Cost of Living | 17. Family Orientation | 28. Liberty | 39. Taxes |
| 7. Crime | 18. Firearms | 29. Market Freedom | 40. Technology |
| 8. Death Rate | 19. Freedom of Speech | 30. Mental Health | 41. Transportation |
| 9. Discrimination | 20. Healthcare Access | 31. Military | 42. Welfare |
| 10. Diversity | 21. Healthcare Cost | 32. Minimum Wage | |
| 11. Education Access | 22. Healthcare Quality | 33. Pollution | |

Version: 3.4.1

2026-04-09

¹Currently developed by a single human modeler, though additional minds would likely improve both the model and the modeler.

► Interpretation Framework

Before jumping down the rabbit hole, I feel it would be wise to provide the reader with context for how each variable is interpreted and structured. To avoid confusion, all variable descriptions in this document will follow a consistent formatting framework. Each section will include a detailed description, methodological scope, and acknowledgment of any known challenges or disputes within the academic literature.

Ontology

Society does not consist of cleanly separable components with absolute borders or dividing lines. In reality, many social phenomena overlap, interact, repel, and co-evolve in extremely complex ways. Because of this, clear conceptual distinctions must be intentionally drawn in order to model them.

For the purposes of Society V1, each variable is assigned abstract boundaries. In the sections that follow, these abstractions are outlined and systematized to minimize conceptual overlap while preserving the ability to model emergent system behavior with reasonable fidelity.

Inclusionary & Exclusionary Scopes

Inclusionary and exclusionary lists define the conceptual perimeter of each variable. These boundaries clarify what is intentionally considered and what is not included within the boundaries of each conceptual variable, ensuring shared interpretation and internal consistency.

This document aims to answer practical questions such as:

- › What does each variable (or their numerical values) actually mean?
 - › What am I changing when I move a variable's value up or down?
 - › How are these concepts measured in the real world?
 - › How are specific influence values determined when research findings vary across studies?
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Research Metrics & Methodology²

All sections will contain the relevant quantitative metrics, secondary indicators, and research approaches corresponding to the verifiable data used to ground each variable's influence weight to the real world.

Model parameters are derived through a structured synthesis of:

- › Established empirical research
- › Cross-national and national statistical datasets
- › Peer-reviewed literature
- › AI tools³
- › Comparative historical analysis

Path-Dependent Influence Weights

Variables are defined broadly, but their output pathways (influence edges) are formulated contextually. Each variable influences others differently, depending on the pathway through which the interaction occurs. In other words, every variable casts different shadows depending on where it lands. Accordingly, each influence edge weight is inferred from the empirical metrics most directly associated with the connection being modeled.

For example, ADDICTION is not simply a singular influence weight applied uniformly across different receptor variables (such as CRIME or LIBERTY). Instead, it functions as a matrix of path-specific influence weights that exert discretely different intensities across its downstream relationships. The ADDICTION variable's influence on DEATH RATE is evaluated through overdose and attributable mortality, while its influence on MENTAL HEALTH is interpreted through the prevalence of dual-diagnosis, surveys, and hospitalization rates (See figure 1 on page 4),

In this way, the model maintains empirical grounding while avoiding the false assumption that a complex societal variable can be represented by a single uniform measurement.

²To find Society V1's citations, datasets, literature, and research sources see: RESEARCH REPOSITORY

³Including platforms such as Consensus AI for literature aggregation and comparative analysis.

1 Addiction

ADDICTION is a variable that encompasses the severity and pervasiveness of two primary dependency domains: Substance Use Disorders and Behavioral Disorders.

Dependency to substances (e.g., alcohol, opioids, stimulants, tobacco) is typically characterized by physiological adaptation; tolerance and withdrawal. In contrast, behavioral addictions (e.g., gambling, sex, pornography, digital dependency) are pattern-reinforced compulsions driven by endogenous reward systems. They lack direct pharmacological dependence and their withdrawal tends to be psychological rather than physiological.

Although the mechanisms between chemical and behavioral dependencies can differ, both produce an equally persistent psychological grip, and both are strongly defined by the phenomena of craving.

Within the domains of substance use and behavioral disorders, there are two separate dimensions: *Physiological Dependency* and *Psychological Compulsion*.

This variable is modeled by placing emphasis on the relevant mechanisms of addiction that propagate into broader societal effects, at the population level, rather than individual experiences at the psychological level.

» ADDICTION includes:

- › Pervasiveness of active addiction
- › Severity and chronicity
- › Recovery and relapse dynamics
- › Economic, judicial, and social burdens
- › Social and institutional degradation

» ADDICTION excludes:

- › General mental illnesses
- › Occasional recreational use
- › Addiction-related mortality
- › Non-compulsive binge behavior
- › Criminality

When the value of ADDICTION increases, it represents a greater prevalence and severity of dependency within the population; conversely, when it decreases, those burdens recede.

In Society V1, ADDICTION functions as both a public health burden and a social destabilizer, influencing crime, productivity, mental health, family stability, and broader societal resilience. ADDICTION is supported by a robust empirical foundation, allowing it to be modeled with a relatively high degree of confidence.

Research Metrics & Methodology

Primary	Secondary
Substance Use Disorder (SUD) prevalence	Self-reported dependency surveys
Overdose rates	Drug-related economic burden estimates
Treatment enrollment rates	Workplace impairment statistics
Relapse rates	Dual Diagnosis Prevalence (psychiatric comorbidity)
Hospitalization Statistics	

Data Challenges & Academic Disputes

- › Under-reporting
- › Inconsistencies in cross-national reporting
- › Measurement gaps
- › Cultural normalization differences
- › Social stigma; dampening treatment and diagnosis visibility
- › Misdiagnosis; also including Over- and Under-diagnosis
- › Criminalization vs public health framing
- › Direction of Socioeconomic causality: (Poverty → Addiction) vs. (Addiction → Poverty)
- › Measurement inconsistency in clinical and survey definitions

The empirical data on addiction could be accurately illustrated with a slice of Swiss cheese. Let's point out some of the holes: For one, surveys cannot record what goes unreported. Bias from Institutional incentives cannot be untangled from those made from financial pressures. Misdiagnoses, whether from inept or complacent professionals, misshape the statistics we are supposed to trust in a significant way. Add to that the inconsistent standards across country-to-country data sharing, and the data we are left with is a distorted picture, a hole, in the very reality it attempts to represent. Therefore, empirical data on addiction = Swiss cheese.

Unfortunately, because of how it manifests, there will always be disputes over whether addiction is more of a public health, criminal justice, or even a genetic issue; which shifts the goalposts for how each case is understood. And it seems that the more standardized a treatment is, the less it works long-term. Our lack of understanding is evident in the growing number of addiction diagnoses, despite the exponential growth of treatment programs and their budgets.

Each row shows the direct effect of increasing this variable on another, where the weight defines both the direction and strength of the influence.

Active Output Edges For: ADDICTION

Output Edges	Weight	Mechanism Summary
AGRICULTURE	-0.08	Small reduction; eroding agricultural productivity and increasing the risk of injury
BIRTHRATE	-0.16	Moderate reduction; diminishing returns become offset by increased unplanned pregnancies
COST OF LIVING	+0.15	Moderate increase; raising healthcare costs, productivity loss, and financial imprudence
CRIME	+0.30	Strong increase; inducing desperation, impaired judgment, and social destabilization
DEATH RATE	+0.31	Heavy increase; actuates more fatal overdoses, accidents, disease progressions, and violence
EDUCATION: ACCESS	-0.07	Small noticeable decrease; reducing enrollment and attendance, increasing disciplinary exclusion
EDUCATION: QUALITY	-0.17	Moderate decrease; impairing concentration, absenteeism, disruption, lower engagement
EMPLOYMENT	-0.23	Substantial reduction; degrading reliability, productivity, and hiring prospects
EQUALITY	-0.11	Slight erosion; concentrating harm in disadvantaged groups which suppresses mobility
FAMILY VALUES	-0.21	Substantial reduction; corroding families, increasing conflict, and diluting inter-generational values
FIREARMS	-0.06	Small decrease; public safety concerns and regulatory pressures limit gun rights
HEALTHCARE: ACCESS	-0.13	Moderate decrease; higher treatment demand, bed saturation, and emergency utilization (capacity)
HEALTHCARE: COST	+0.32	Heavy increase; expenses from ER visits, inpatient care, chronic disease, and long-term treatments
HEALTHCARE: QUALITY	-0.15	Moderate decline; overload and case complexity reduces capacity and care effectiveness
HOMELESSNESS	+0.18	Moderate increase; financial instability and social breakdown raises the risk of housing loss
IMMIGRATION	-0.04	Negligible effect; slightly constrains admissibility and retention through policy filters
INNOVATION	-0.16	Moderate suppression; impairing focus, consistency, and sustained problem-solving
LAW ENFORCEMENT	+0.20	Substantial increase; more frequent overdose response, civil-disorder, and drug-related crime
LIBERTY	-0.22	Substantial decrease; withdrawal and compulsive behaviors erode autonomy and self-directed choice
MARKET FREEDOM	-0.11	Slight decrease; Undermining agency, while exploitative incentives distort voluntary exchange
MENTAL HEALTH	-0.35	Heavy decrease; reinforcing depression, anxiety, and stress disorders through comorbid feedback
MILITARY	-0.16	Moderate reduction; narrowing recruitment eligibility, readiness, and operational performance
POVERTY	+0.24	Substantial increase; reduces employability and degrades financial management
PUBLIC TRUST	-0.12	Slight decrease; stigma and perceived unpredictability reduce interpersonal trust
RULE OF LAW	-0.22	Substantial decrease; inconsistency, ineffectiveness, and perceived illegitimacy reduces integrity
SICKNESS/DISEASE	-0.18	Moderate decrease; eroding serenity and clarity, reducing spirituality to superficial utilitarian states
SPIRITUALITY/GOD	+0.32	Heavy increase; inflicting physiological harm, comorbidities, and general health burdens
WELFARE	+0.16	Moderate increase; feeds codependency and reliance while pressuring expansion of welfare support

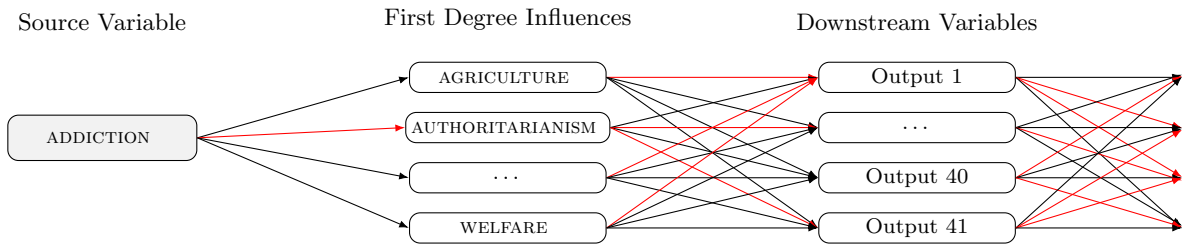


Figure 1: Illustration of how a variable, such as ADDICTION, projects influence across multiple downstream nodes. Each edge carries a unique weight and shaping function, transforming the signal as it propagates through the network. Red edges indicate zero direct influence; black edges indicate active influence (non-zero weight). Selected outputs extend further, projecting their own downstream networks.

2 Agriculture

In Society V1, AGRICULTURE looks at much more than the total output of a seasonal harvest. Although it does take into account capacity and the same types of metrics hedge-fund managers love to illustrate in commodity charts, it reaches out across multiple agricultural domains.

Within the model, it is treated as a dense system variable. It aggregates the full agricultural ecosystem: fields and livestock, management and labor, machines and ecological sustainability, cross-sector coordination, soil, logistics, knowledge, capital dynamics, technological adoption, and the living choreography that keeps agricultural products flowing through store shelves and onto the tables of hundreds of millions of families around the world. It represents a picture of the quiet mother of civilization itself.

In practice, this means agriculture in the model is shaped by real pressures, labor availability, immigration, land use/property constraints, cost structures, logistics, and policy. These forces don't just influence output; they shape how stable the system is, how efficiently it operates, and how it responds to other variables. The weights assigned to AGRICULTURE are not arbitrary, they are inferred from how these pressures consistently show up in real-world data and how strongly they interact with the rest of the system.

» AGRICULTURE includes:

- › Productive capacity of crops and livestock
- › Quality of labor within agricultural systems
 - Agricultural workforce availability
 - Health, skill level, and productivity
- › Mechanization, tools, and technological integration
- › Infrastructure and logistical reliability
- › Environmental Sustainability and soil vitality
- › Community-level agricultural cohesion and knowledge transmission

» AGRICULTURE excludes:

- › Broader industrial output
- › Climate dynamics
- › National wealth outside agricultural systems
- › Agricultural regulations, laws, or policies

Agriculture is a foundational stabilizer in the model because it is a foundational stabilizer in reality. When agricultural systems are strong, food security, economic stability, rural communities, and public health tend to strengthen with it. When agriculture weakens, fragility spreads outward in quiet but compounding ways.

Research Metrics & Methodology

Primary	Secondary
Crop yield stability	Input efficiency (fertilizer, energy, water, etc.)
Livestock productivity	Income stability
Food affordability pressure	Post-harvest loss rates
Agricultural labor productivity	Soil health and land degradation
Production-storage-distribution continuity	Import dependence and commodity-shock exposure

Data Challenges

- › Subsistence and informal production are often under-counted
- › Cross-country reporting standards are inconsistent
- › Climate shocks can obscure structural agricultural performance
- › Policy and subsidy regimes can distort market-facing indicators
- › Many effects are lagged, so short windows can misrepresent direction and strength

These limitations mean Agriculture should be interpreted as a systems variable, not a single-output statistic. Short-term output can improve while structural resilience weakens, or vice versa.

Known Confounds & Academic Disputes

- › High-output industrial models vs long-run ecological sustainability
- › Scale efficiency vs resilience of distributed/local food systems
- › Direction of causality in development loops (Poverty:low → Agriculture vs. Agriculture → prosperity:low)
- › Institutional quality and infrastructure acting as hidden mediators in social outcome pathways

Because research does not always agree, the influence values in Society V1 are not copied from any single study. They are informed by broader mechanisms, consistent directional signals, and the clearest patterns that appear across multiple sources. As better evidence emerges, the values can be adjusted and refined.

Active Output Edges For: AGRICULTURE

Output Edges	Weight	Mechanism Summary
ADDICTION	+0.05	Negligible increase; empirical evidence shows that pain, strain, and instability raises addiction risk
AUTHORITARIANISM	+0.10	Small increase; centralized land control and resource dependency feeds authoritarian traits
BIRTHRATE	+0.07	Very small increase; food stability and family labor incentives enable larger families
COST OF LIVING	-0.12	Slight decrease; increases in local food supply reduces reliance on external markets
CRIME	-0.14	Slight decrease; food and supply stability reduces scarcity pressure and economically driven crime
DEATH RATE	-0.17	Moderate decrease; nutritional surplus improves survivability, offset by occupational fatalities
DISCRIMINATION	+0.10	Small increase; land, credit, and institutional gate-keeping reinforce exclusion
EMPLOYMENT	+0.21	Substantial increase; agricultural production expansion directly increases labor demand
ENERGY PRODUCTION	+0.18	Moderate increase; output expansion drives both farm energy demand and bio-energy supply
EQUALITY	-0.09	Small decrease; asset-dependent participation can amplify emergent unequal outcomes
FAMILY VALUES	+0.14	Slight increase; agrarian structures reinforce family-centered norms across society
FIREARMS	+0.10	Small increase; rural utility and cultural spillover slightly raise firearm prevalence
FREEDOM OF SPEECH	-0.08	Small decrease; grassroots freedoms are offset by industrial-scale incentives for information control
HEALTHCARE: ACCESS	+0.06	Small increase; economic stability improves engagement but constrained by rural access limits
HEALTHCARE: COST	+0.05	Negligible increase; risk exposure raises costs despite indirect health improvements
HEALTHCARE: QUALITY	+0.03	Weak increase; economic support offsets rural access limitations and system strain
HOMELESSNESS	-0.06	Small decrease; rural stability and reduced poverty slightly lowers displacement risk
IMMIGRATION	+0.13	Slight increase; agricultural labor demand creates a meaningful pull for migrant workers
INFRASTRUCTURE	+0.17	Moderate increase; agriculture creates demand and justification for infrastructure expansion
INNOVATION	+0.15	Moderate increase; efficiency pressure drives applied agricultural innovation demand
LIBERTY	+0.03	Negligible increase; decentralization and rural autonomy support cross-cultural freedoms
MENTAL HEALTH	-0.12	Slight decrease; stress, isolation, and economic instability outweigh limited lifestyle benefits
POLLUTION	+0.31	Heavy increase; fertilizer runoff, livestock waste, and emissions pollute air, soil, and water
POVERTY	-0.28	Strong decrease; productivity, employment, and economic gains reduce poverty
PUBLIC TRUST	+0.04	Negligible increase; food stability and rural cooperation slightly reduce uncertainty
SICKNESS/DISEASE	+0.18	Moderate increase; emissions, pathogens, and particulates elevate disease burdens
SPIRITUALITY/GOD	+0.25	Strong increase; dependence on natural cycles drives ritualization and organized spiritual systems
TECH	+0.18	Moderate increase; agricultural pressures more innovation, especially in efficiency and optimization
TRANSPORTATION	+0.29	Strong increase; agriculture is a primary driver for transport expansion, tapering at high levels
WELFARE	-0.12	Slight decrease; food affordability and rural income marginally reduces welfare program demand

words

Active Output Edges: (SLIDER HERE)

[illegible]

3.1 Birthrate

description here

3.2 Corruption

description here

3.3 Cost of Living

description here

3.4 Crime

description here

3.5 Death Rate

description here

3.6 Discrimination

description here

3.7 Diversity

description here

3.8 Education Access

description here

3.9 Education Cost

a

3.10 Education Quality

a

3.11 Employment

a

3.12 Energy Production

a

3.13 Equality

a

3.14 Family Orientation

a

3.15 Freedom of Speech

a

3.16 Firearms

a

3.17 Healthcare Access

a

3.18 Healthcare Cost

a

3.19 Healthcare Quality

a

3.20 Homelessness

a

3.21 Immigration

includes legal retention / admissibility / long-term settlement

- 3.22 Infrastructure
 - a
- 3.23 Innovation
 - a
- 3.24 Law Enforcement
 - a
- 3.25 Liberty
 - a
- 3.26 Market Freedom
 - a
- 3.27 Mental Health
 - a
- 3.28 Military
 - a
- 3.29 Minimum Wage
 - a
- 3.30 Pollution
 - a
- 3.31 Poverty
 - a
- 3.32 Public Trust
 - a
- 3.33 Rule of Law
 - a
- 3.34 Sickness/Disease
 - a
- 3.35 Spirituality/God
 - a
- 3.36 Taxes
 - a
- 3.37 Technology
 - a
- 3.38 Transportation
 - a
- 3.39 Welfare Support
 - a

words

Active Output Edges: (AUTHORITARIANISM)

[illegible]